

How Many In-Context Examples Is a Task Description Worth?

Bruce Quan Nguyen

[Affiliation]

Draft of 1 October 2026

Abstract

Theories of in-context learning (ICL) model a prompt’s examples, not its task description. We ask how many examples a description is worth. In a Bayesian reading of ICL both are data about a latent task. In in-context linear regression, a description’s effective sample size (ESS) is how many examples lower a Bayes-optimal predictor’s one-step regret as much as it does. Its precision ratio r is the fraction of prior variance it leaves, its reliability p the probability that it is correct. A reliable description is worth about $d(1 - r) + 1/(ra_0)$ examples in d dimensions, with a_0 the prior signal-to-noise ratio: its high-SNR worth plus the long-run worth that remains once examples accumulate. Reliability caps precision only while nothing can check the description. Alone, an unreliable description keeps at least $1 - p$ of an empty prompt’s regret, so its ESS saturates: at $p = 0.9$ in sixteen dimensions, a hundredfold gain in precision adds ten examples, a hundred when reliable. Examples verify it, and its worth rises toward a fraction p of a reliable description’s. Small meta-trained Transformers reproduce the Bayes-optimal ESS of a description alone, except that one trained on precise unreliable descriptions under-values them; one trained only on reliable descriptions trusts an unreliable one fully, so a precise one is worth almost nothing. Our claims concern Bayes-optimal predictors and small meta-trained sequence models, not large language models.

1 Introduction

In-context learning (ICL) lets a model adapt to a new task from a prompt (Brown et al., 2020), which usually pairs a natural-language task description with a few demonstrations; formal theories of ICL model only the demonstrations (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022). The evidence on descriptions is mixed: a good one can be worth hundreds of data points (Le Scao and Rush, 2021; Honda et al., 2025), a misleading one can perform about as well as an informative one (Webson and Pavlick, 2022), and enough examples override an explicitly stated bias (Gupta et al., 2025). Theoretical models with a descriptor (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026) do not put its value on the same scale as the examples.

We quantify a task description’s value as the number of in-context examples it can replace. In the Bayesian reading of ICL (Xie et al., 2022), the pretraining distribution is the prior over a latent task and everything in the prompt is data: the examples observe the task indirectly through inputs, the description directly, if imperfectly. Conditioning on the description first gives the prior under which the examples are read, which is where the phrase “the description is the prior” comes from. Bayesian statistics weighs such information by its *effective sample size* (ESS) (Morita et al., 2008), which depends on how concentrated it is and whether it agrees with the truth (Evans and Moshonov, 2006): for a task description, its **precision**, how far it reduces posterior uncertainty, and its **reliability**, the probability that it is correct.

Contributions.

1. In noisy in-context linear regression (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where the Bayes-optimal predictor is exact, we define a description’s ESS by matching the predictor’s one-step log-loss regret given the description to that given n examples, with or without further examples in hand (§3).
2. An unreliable description induces a hard floor on regret, set by its error probability $1 - p$ and its entropy $H(p)$, so its ESS saturates however precise it is. Examples verify the description and lift the cap: its worth with n examples in hand rises with n toward a fraction p of a reliable description’s, whose own worth falls (§5).

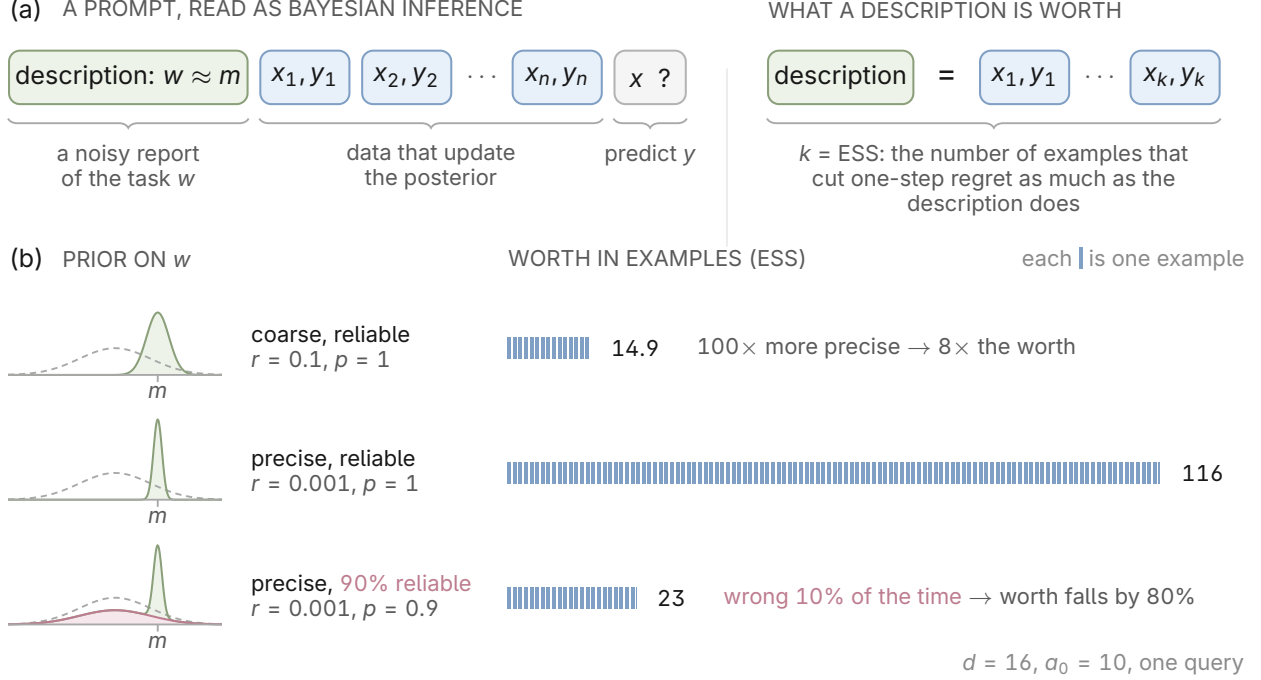


Figure 1: **A task description is worth a number of in-context examples, set by its precision and capped by its reliability.** (a) We read a prompt as Bayesian inference over a latent task w under the pretraining prior: the description is a direct noisy observation of w and the examples are indirect ones. The description’s effective sample size (ESS) is the number of examples k that lowers a Bayes-optimal predictor’s one-step regret as much as the description does. (b) Three descriptions and their ESS. The precision ratio r sets how narrowly the prior concentrates around the stated m ; reliability p is the chance the description is correct, and otherwise w follows the base prior (dashed; its share $1 - p$ shaded pink). Prior shapes are sketches; ESS values are from Table 3 in Appendix A ($d = 16, a_0 = 10$).

3. A perfectly reliable description is worth $\text{ESS} \approx (d - 1)(1 - r) + 1/(ra_0)$ examples, with d the dimension, r its precision ratio, and a_0 the prior SNR: its high-SNR worth plus its long-run worth, within two examples across the grid (§4).
4. Small Transformers meta-trained on this process, one per setting, reproduce the Bayes-optimal ESS of a description alone in three of four settings and under-value a precise unreliable description in the fourth; trained only on reliable descriptions, they do not discount an unreliable one (§6).

2 Related Work

Bayesian Perspectives on ICL. ICL can be framed as implicit Bayesian inference (Xie et al., 2022). In the context of linear regression, meta-trained Transformers effectively implement Bayes-optimal ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024). Genewein et al. (2025) score meta-trained predictors by log-loss regret against the task oracle and compare the curve with the Bayes-optimal predictor’s; we adopt both. Zhu et al. (2026) show that a Transformer given a prefix of related datasets performs amortised hierarchical Bayesian prediction, matching the oracle across prior families, which is the closest demonstration that a prior-carrying prefix is used Bayes-optimally; they do not price the prefix in examples or model an unreliable one. None of these works translate the influence of a description into an equivalent number of in-context examples.

Task Descriptions in ICL Theory. Recent studies have begun exploring task descriptions theoretically. Huang and Ge (2025) include a descriptor token carrying the input mean; it helps their learner by centring the inputs but carries no information about the weights, so to a Bayes-optimal predictor it is worth zero examples. Lin et al. (2025) guide

models using hypothesis classes, and Tong et al. (2026) show that a prediction’s dependence on the instruction, right or wrong, decays exponentially in the number of demonstrations. Lin and Lee (2024) analyze Gaussian-mixture task priors whose components are fixed by pretraining; ours has two components centred by the description. None of these works quantify the description’s value in terms of sample size.

Effective Sample Size. In statistics, Morita et al. (2008) define the ESS of a prior by matching its information to that of a baseline prior plus a sample; Reimherr et al. (2021) make the prior sample size a function of the data in hand, $M(k)$, and leave prediction-based uncertainty measures as an open direction; Neuenschwander et al. (2020) note that a mixture prior’s ESS is not the weighted average of its components’. Reznik (2026) defines a borrowed ESS under predictive log loss for a power prior, with a divergence between historical and current data in the role our $1 - p$ plays; it has no mixture and does not depend on the current data. Our ESS is the prior sample size of Reimherr et al. (2021) with one-step predictive regret as the uncertainty measure, so it depends on the examples in hand. Our model of reliability is the robust mixture prior of Schmidli et al. (2014), used in clinical trials to incorporate possibly conflicting historical data: their worked example puts weight 0.1 on a vague component, which is our $p = 0.9$, and our base prior plays the vague component’s role.

3 Problem Formulation

Task Distribution. We consider tasks defined by a weight vector $w \in \mathbb{R}^d$. An in-context example consists of an input $x \sim \mathcal{N}(0, I_d)$ and a corresponding label $y = w^\top x + \epsilon$, where the noise is $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. The base prior over tasks is $w \sim \mathcal{N}(0, s_0^2 I_d)$. For simplicity, we set $\sigma_y = 1$ and define the prior signal-to-noise ratio (SNR) as $a_0 = s_0^2 / \sigma_y^2$. We take $a_0 = 10$, above the values of 1 to 4 in the noisy settings of Garg et al. (2022); Raventós et al. (2023); Panwar et al. (2024) and within the range swept by Akyürek et al. (2023), so that a precise description can be worth many examples.

Task Description Model. We model a task description as a vector $m \in \mathbb{R}^d$ that provides information about w . The description’s *precision ratio* is $r \in (0, 1)$, meaning the variance around the description is $s_\ell^2 = r s_0^2$. The description is generated hierarchically via $m \sim \mathcal{N}(0, (1 - r) s_0^2 I_d)$, ensuring the marginal distribution of w matches the base prior. A description is not always correct. We define its *reliability* p as the probability that the description is accurate. If correct, $w \sim \mathcal{N}(m, s_\ell^2 I_d)$; if incorrect (with probability $1 - p$), w simply follows the base prior $\mathcal{N}(0, s_0^2 I_d)$ independently of m . A wrong description is thus unrelated to the task, as in robust mixture priors, rather than deliberately misleading: a description that pointed away from w by rule would be information about w to a learner that knew the rule, and harm only a learner that did not. Consequently, the Bayes-optimal prior conditioned on the description is a robust mixture (Schmidli et al., 2014): $p \mathcal{N}(m, s_\ell^2 I_d) + (1 - p) \mathcal{N}(0, s_0^2 I_d)$. Equivalently, m is a noisy observation of the weights themselves, $m \mid w \sim \mathcal{N}((1 - r)w, r(1 - r)s_0^2 I_d)$ when correct, so the description is data about w and the ESS is an exchange rate between direct observations of w and indirect ones through x . A description with precision ratio r locates each weight to within $\sqrt{r}$ of its typical size s_0 : 32% at $r = 0.1$, 10% at $r = 0.01$, 3% at $r = 0.001$. The formulation treats the description as a numerical observation rather than text to be parsed.

Regret and Effective Sample Size. We score predictors by one-step predictive log loss for a new query x . The regret is the expected excess log loss over an oracle that knows w exactly, the expectation taken over tasks, inputs, and labels: the per-step version of the regret of Genewein et al. (2025). Log loss, rather than the squared error of Garg et al. (2022), is a proper scoring rule and the loss in which the mixture bound of §5 holds. The prior sample size of Reimherr et al. (2021) measures a prior π against a baseline π_b given data I by the M that solves $U_{\pi_b}(I + M) = U_\pi(I)$: the baseline with M further observations is as uncertain as the prior with the data. We take the base prior as π_b , the description-conditioned prior as π , the n examples as I , and regret as the uncertainty U .

Definition 1 (Worth of a description). *Fix p and r . Let $R^{\text{desc}}(n)$ be the regret of the Bayes-optimal predictor given the description and n examples, and $R^{\text{ex}}(n)$ its regret given n examples and no description. The worth of the description with n examples in hand is $\text{ESS}_n = n^* - n$, where n^* (interpolated linearly between integers) solves $R^{\text{ex}}(n^*) = R^{\text{desc}}(n)$: the further examples an examples-only predictor needs to match it, the prior sample size $M(n)$ above, which is negative when the description hurts. The effective sample size of the description is its worth alone, $\text{ESS} = \text{ESS}_0$.*

The ESS compares prompts of two kinds, the description alone against examples alone, and asks what the description is worth on its own in the currency of examples on their own. ESS_n asks the same with n examples already present; §5 shows why that matters for unreliable descriptions. Since we compare Bayes-optimal predictors under different conditioning sets, the ESS measures information content rather than the specific architecture of a learner.

4 The Value of Precision

We first analyze a perfectly reliable description ($p = 1$). Given n examples $X_n \in \mathbb{R}^{n \times d}$, the posterior covariance under a Gaussian prior with SNR a is $\Sigma = \sigma_y^2(I/a + X_n^\top X_n)^{-1}$. This is independent of the examples' labels and the description m . The expected one-step regret simplifies to $\frac{1}{2}\mathbb{E} \log(1 + x^\top \Sigma x)$. A reliable description equates to $n = 0$ and $a = ra_0$, whereas relying solely on examples uses $a = a_0$.

Proposition 1 (Worth of a reliable description). *Let ψ be the digamma function and $c = 1/(ra_0)$, the long-run worth of the description (§5).*

- (i) Coarse descriptions, $c \leq 1$. For $n < d$, $R^{\text{ex}}(n) \geq \frac{1}{2} [\log(2a_0) + \psi(\frac{d-n}{2})]$, with equality as $a_0 \rightarrow \infty$, and as $rd \rightarrow \infty$, uniformly in $c \leq 1$,

$$\text{ESS} = (d-1)(1-r) + (1-r)c + O\left(\frac{1}{rd}\right).$$

- (ii) Precise descriptions, $c \geq d$. For $n \geq d$, $R^{\text{ex}}(n) \leq \Phi(n) := \frac{1}{2} [\psi(\frac{n+1}{2}) - \psi(\frac{n-d+1}{2})]$, with equality as $a_0 \rightarrow \infty$, and as $c/d \rightarrow \infty$, uniformly in $a_0 \geq 1$,

$$\text{ESS} = c + d + 1 - \frac{1}{a_0} + O\left(\frac{d}{c}\right).$$

In every setting, $\text{ESS} \leq c + d + 2$.

Idea. paragrah =====

Idea. Regret is half the log of the leftover variance in the prediction, and a description with precision ratio r leaves $ra_0 = 1/c$ of the noise variance in each of d directions. Examples remove variance in two ways. The first $n < d$ of them each kill one direction outright at high SNR, so the leftover is a_0 times a χ_{d-n}^2 ; matching it to the description's $ra_0\chi_d^2$ gives $d - n \approx rd$, the high-SNR term $(d-1)(1-r)$. Beyond $n = d$ every direction is seen and the leftover is χ_d^2/χ_{n-d+1}^2 , with mean $d/(n-d-1)$; matching it to d/c gives $n \approx c + d + 1$. The prior itself is worth $1/a_0$ examples, which is the $-1/a_0$ in (ii) and, after the factor $1-r$, the $c - 1/a_0$ in (i). The proof, with the error terms, is in Appendix B.

Both regimes say the same thing: the worth of a description alone is its high-SNR worth plus its long-run worth, $\text{ESS} \approx (d-1)(1-r) + c$, within two examples. In (i) the long-run term enters as $(1-r)c$, which is less than c by $1/a_0 < 1$; in (ii) the exact constant is $c + d + 1 - 1/a_0$, which exceeds $(d-1)(1-r) + c$ by $2 - 1/a_0 + (d-1)r \in [1, 2)$ up to the stated error. The two regimes meet at $c = 1$, where the description pins w down as tightly as the label noise does, and at $c \approx d$ its worth passes d : below that, each unit of precision removes directions and the worth stays under d ; above it, every direction is already seen and the worth grows like c , the examples needed to match the description's variance, plus the $d + 1$ that d noisy examples cannot supply, minus the $1/a_0$ examples the prior is already worth. The universal bound $\text{ESS} \leq c + d + 2$ holds everywhere, including the crossover $1 < c < d$ where neither expansion applies. On the grid of Table 2 and Figure 2(a), the additive rule is within 2.2 examples of the exact ESS in every cell with $a_0 \geq 10$; at $d = 64$, $a_0 = 100$, $r = 0.1$ the exact ESS is 56.85 against 56.79 from (i), and at $d = 16$, $a_0 = 10$, $r = 0.001$ it is 116.8 against 116.9 from (ii).

In words: a description that removes nine tenths of the prior variance is worth nine tenths of the dimension plus its long-run worth of one example, 14.9 at $d = 16$ and 57.7 at $d = 64$; one that leaves a thousandth is worth its long-run 100 plus $d + 1$, 117 at $d = 16$. Table 2 (Appendix A) lists the exact values and Figure 2(a) traces the curve at $d = 16$.

5 The Bottleneck of Reliability

In reality, descriptions can be flawed or ambiguous. With an unreliable description the Bayes-optimal predictive is a mixture of two components, one that trusts the description and one that ignores it, weighted by the posterior probability that the description is correct. The following bound measures what the hedge costs, with and without examples to check the description against.

Proposition 2. Let $R^{\text{desc}}(n)$ be the regret of a description with reliability p and precision ratio r with n examples in hand, $R_{p=1}^{\text{desc}}(n)$ the same for a reliable one, $R^{\text{ex}}(n)$ the regret with n examples and no description, and π_n the Bayes-optimal predictor's posterior probability that the description is correct after n examples. For every $n \geq 0$, $p \in (0, 1)$, and $r \in (0, 1)$,

$$p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n) \leq R^{\text{desc}}(n) \leq p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n) + \mathbb{E}[-\log \pi_n(Z)],$$

where Z indicates that the description is correct and $\pi_n(Z)$ is the posterior weight on the true component. The identification term $\mathbb{E}[-\log \pi_n(Z)]$ equals $H(p)$, the binary entropy in nats, at $n = 0$ and is non-increasing in n . In particular, alone ($n = 0$) the regret is at least $(1 - p) R^{\text{ex}}(0)$ for every r , so the ESS cannot exceed the n at which $R^{\text{ex}}(n) = (1 - p) R^{\text{ex}}(0)$, and as $r \rightarrow 0$ the regret lies between $(1 - p) R^{\text{ex}}(0)$ and $(1 - p) R^{\text{ex}}(0) + H(p)$.

Idea. Compare the learner with a twin who is also told whether the description is correct. The twin's regret is $p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n)$. The learner cannot beat the twin, since extra information never hurts a Bayes-optimal predictor. And its predictive puts weight π_n on the twin's answer, so it loses at most $-\log \pi_n(Z)$: with no examples, $\log(1/p)$ nats when the description is correct and $\log(1/(1 - p))$ when it is wrong, which average to $H(p)$; more examples can only sharpen the posterior on Z in expectation. The upper bound is the classical mixture bound (Cesa-Bianchi and Lugosi, 2006, Cor. 3.1 and §9.2) (Merhav and Feder, 1998, §V) applied to the two components, and the lower bound is the Bayes-optimality of the informed twin; what is new is what they say about the ESS. The proof is in Appendix C.

Alone. With no examples, nothing can identify the correct component: the learner hedges by $1 - p$ and pays up to $H(p)$ for not knowing which case it is in, however precise the description. For $d = 16$ and $a_0 = 10$, $R^{\text{ex}}(0) = 2.51$ nats and the reliable description's regret at $r = 0.001$ is 0.07; at $p = 0.9$ the two bounds are 0.32 and 0.64, and the computed regret is 0.57. So an unreliable description can replace only a bounded number of examples, and fewer than the floor allows: at $p = 0.9$ the floor's cap is 40 examples, but the identification cost is all of $H(p)$ in the precise limit, because a description that pins w exactly or is simply wrong leaves nothing to tell the two cases apart, and the ESS saturates at 25 as $r \rightarrow 0$ (dotted in Figure 2); at $p = 0.99$ the cap is 325 and the limit 135. Figure 2 and Table 3 (Appendix A) show the saturation across precision. For a perfectly reliable description, refining r from 0.1 to 0.001 raises the ESS from 14.9 to 116; at $p = 0.9$ it only grows from 13.1 to 23.1, while the reliable description's keeps rising. Even a 1% unreliability ($p = 0.99$) nearly halves the ESS of the most precise description, 64 against 116, while leaving the coarse one almost untouched: the curve stays within a tenth of the reliable one down to $r = 0.01$. The same saturation patterns manifest at $d \in \{4, 64\}$ and $a_0 = 100$.

With examples in hand. Examples change the picture: they can reveal whether the description is correct, and once they have, the predictor can use its full precision. In Proposition 2 the floor $(1 - p) R^{\text{ex}}(n)$ decreases with n and the identification term cannot rise, so the cap on the description's worth rises as examples accumulate. How far it rises has a simple form once n is large enough that the examples have identified the description and $R^{\text{ex}}(n) \approx d/(2n)$. Identification requires the two components to be distinguishable: the identification term tends to the residual entropy of Z given w and m , which is negligible here because the components are far apart (their KL divergence is of order $\frac{d}{2} \log(1/r)$) but stays positive for a coarse description, 0.17 nats at $d = 5$, $r = 0.5$. Writing $c = 1/(ra_0)$, the reliable-description regret is then about $d/(2(n + c))$, so the floor is $\frac{d}{2} [\frac{p}{n+c} + \frac{1-p}{n}]$, and solving $d/(2n^*)$ equal to it gives

$$\text{ESS}_n \approx \frac{n p c}{n + (1 - p) c} \xrightarrow{n \rightarrow \infty} p c = \frac{p}{r a_0},$$

against $\text{ESS}_n \rightarrow c = 1/(ra_0)$ for a reliable description. So a precise unreliable description, once examples have verified it, is worth p times what a reliable one is worth; alone, the $n = 0$ case caps it far lower. The formula applies only once n exceeds d , and the exact learner shows two regimes before that limit (Figure 2(c); $r = 0.001$, $p = 0.9$, $d = 16$, $c = 100$). The first one or two examples identify the description: the identification term falls from $H(p) = 0.33$ nats to 0.07 after one example and 0.02 after two, and the worth rises from 23 examples alone to 30 after one. It then stays near 30 until the examples outnumber the dimension, because the wrong-description branch still carries the full examples-only regret, and rises once $n > d$ as that branch's regret shrinks: 29 at $n = 10$, 37 at 16, 88 at 100, against the formula's 82 at $n = 100$ and a limit of 90. The reliable description drifts from 117 to 108 over the same range, and at $r = 0.01$, where $c = 10$ is below d , the unreliable description gains one example after the first example and then loses

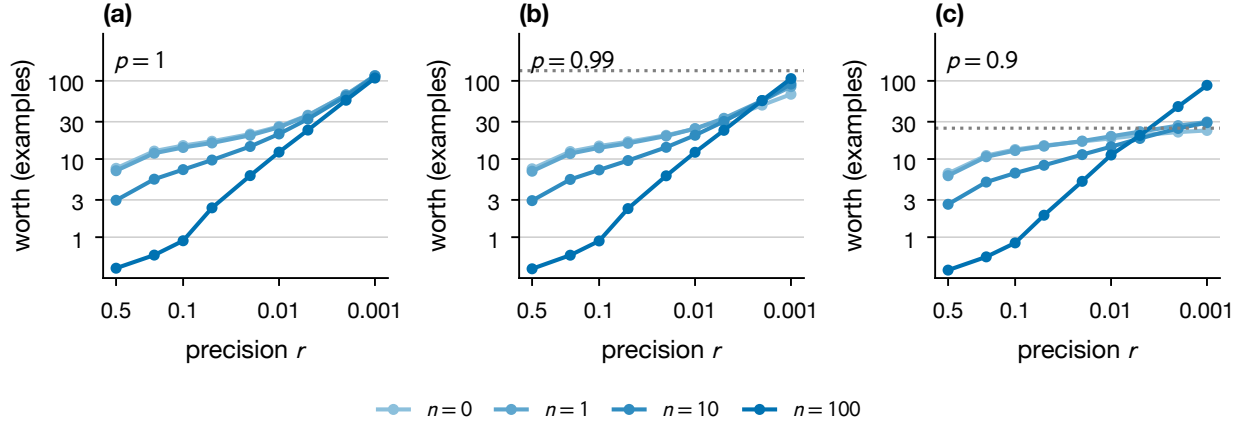


Figure 2: **Reliability caps precision only while nothing can check the description.** Worth of a description against its precision with n examples in hand (ESS_n of Definition 1), one panel per reliability; $d = 16$, $a_0 = 10$, log scales, one seed of 8,000 prompts. Dotted: the limit of the $n = 0$ ESS as $r \rightarrow 0$ at $p = 0.9$, 25 examples. Alone, the $p = 0.9$ description saturates at 23 examples while the reliable one keeps rising; with 100 examples in hand the three panels nearly coincide (109, 107, 88 at $r = 0.001$).

worth like the reliable one. Figure 2 shows this across precision: with 100 examples in hand the most precise description is worth 88 examples at $p = 0.9$ against 109 when reliable, where alone it was worth 23 against 117. Reliability caps precision until examples can check the description, and then costs a factor p .

A fully trusting learner. The cap and its lifting assume a learner calibrated to p . A learner that takes the description as certainly correct, trust $q = 1$ when the truth is $p = 0.9$, pays for the wrong tenth in full: its predictive is as tight as the description, so when the description is wrong the answer lies far outside it. At $d = 16$, $a_0 = 10$, the description alone costs this learner 2.2 nats at $r = 0.1$, where no description costs 2.51, so a coarse description still helps (6.5 examples against 13.2 when calibrated). At $r = 0.01$ it costs 6.3 nats and at $r = 0.001$ 13.4: worse than no description at all. Examples repair this only slowly, because the wrong tenth carries a prior as tight as the description’s: after 100 examples the $r = 0.01$ description still leaves the learner 58 examples behind one that had no description, and the $r = 0.001$ description leaves 4.0 nats of regret against 0.09 without it. To a learner that cannot doubt the description, precision is a liability. §6 shows this learner trained: a network that has never seen a wrong description behaves like it.

6 Meta-Trained Transformers

Design. We meta-train Transformers on prompts from the process of §3 and compare them with the Bayes-optimal predictor on the same prompts, following Genewein et al. (2025). Prompts use the prefix embedding of Huang and Ge (2025): an optional descriptor token carrying m , up to 15 example tokens each holding an input beside its answer, and a final query token, with two indicator coordinates marking the token type; there is no positional encoding, so the examples are exchangeable. Architecture and optimisation follow Garg et al. (2022): 12 layers, 8 heads, width 256, Adam at 10^{-4} , fresh prompts at every step. The network outputs a mean and a log variance for the query’s answer and is trained on the Gaussian log loss, so its regret is on the scale of §3. We set $d = 5$ and $a_0 = 10$ (the network sees $w \sim \mathcal{N}(0, I)$ and noise variance 0.1, the same task in the units of Garg et al. (2022)), draw the number of examples uniformly from 0 to 15, and omit the descriptor in half the prompts. The descriptor supplies only m ; each network is trained at one r and one p , which it must learn from the training distribution. Each model trains for 40,000 steps at batch 1024 without a curriculum (Garg et al. (2022) use batch 64, 500,000 steps, and a curriculum over dimension and prompt length), one per setting; in a pilot the mean gap to the Bayes-optimal predictor fell from 0.07 nats at 5,000 steps to 0.03 at 20,000 and about 0.02 from 30,000 to 40,000.

Table 1: Networks against the Bayes-optimal predictor ($d = 5$, $a_0 = 10$, one trained model per row, 20,000 evaluation prompts). “Trained” is the Bayes-optimal predictor with trust equal to the training reliability p_{train} , the predictor each network is trained toward; “calibrated” is the predictor with trust equal to the test reliability, shown where the two differ. The last two columns are the mean regret gap, network minus the “trained” predictor, over $n = 0, \dots, 15$, with and without a descriptor.

r	p_{train}	p_{test}	ESS			Regret gap (nats)	
			Network	Trained	Calibrated	Descriptor	None
0.05	1	1	6.69	6.68	—	0.023	0.022
0.5	1	1	2.25	2.26	—	0.019	0.019
0.05	0.9	0.9	3.94	4.98	—	0.070	0.025
0.5	0.9	0.9	1.82	1.85	—	0.019	0.016
0.05	1	0.9	0.72	≤ 0	5.04	0.000	0.022
0.5	1	0.9	1.64	1.72	1.85	0.024	0.019

Readouts. We evaluate each network on 20,000 fresh prompts for every n from 0 to 15, with and without a descriptor. As in Definition 1, the ESS is where the network’s regret with the descriptor alone meets its own regret curve for examples alone. The standard error of each regret is below 0.01 nats, which moves an ESS by less than 0.1 examples. We also evaluate the two models trained at $p = 1$ on prompts whose descriptors are correct only with probability 0.9.

Results. For reliable descriptors the networks reproduce the Bayes-optimal values (Table 1, Figure 3): ESS 6.7 against 6.7 for the precise descriptor and 2.25 against 2.26 for the coarse one, with regret about 0.02 nats above the Bayes-optimal predictor’s on average over n . The coarse unreliable descriptor is also reproduced, 1.83 against 1.85. The precise unreliable descriptor is not: the network’s ESS is 3.9 against 5.0, its regret from the descriptor alone is 0.3 nats above the Bayes-optimal predictor’s, and its implied trust is 0.85 rather than 0.9, where the implied trust at n examples is the q whose Bayes-optimal predictor has the posterior-mean prediction closest, in mean squared difference over prompts, to the network’s. The gap closes as examples arrive (Figure 3c). It is not under-training: continuing the same model to 80,000 steps on fresh prompts leaves its ESS at 3.9 and its implied trust at 0.85. With one trained model per setting we report it as an observation rather than a finding.

Transfer to unreliable descriptors. The models trained at $p = 1$ behave like a Bayes-optimal predictor that trusts the descriptor fully. With one descriptor in ten wrong, the model trained on precise reliable descriptors gets almost nothing from a descriptor: ESS 0.7 where a calibrated predictor would get 5.0, and implied trust 0.99. The fully trusting Bayes-optimal predictor’s regret with the descriptor is no lower than without one (1.869 against 1.866 nats, within Monte Carlo error), so its ESS is about zero (≤ 0 in Table 1). The coarse model is forgiven its trust, 1.6 against the calibrated 1.9, because a wrong coarse descriptor does little harm. This is the fully trusting learner of §5 seen trained: the cost of unreliability falls on precise descriptors, and a learner that has never seen a wrong one pays it in full.

7 Discussion

For a Bayes-optimal predictor, a task description is worth a definite number of in-context examples. When the description is reliable, its worth is the share of the dimension it pins down plus its long-run worth, the examples that would match its variance. Otherwise the predictor must hedge against its being wrong, so a very precise description’s worth saturates. The hedge is forced only while nothing can check the description; a few examples lift it, and a precise unreliable description is then worth more than alone. Reliability is not a fixed discount but a cost that examples pay down, to a factor p . This holds for a learner calibrated to the description’s reliability; to one that cannot doubt it, a precise description that is sometimes wrong is worse than none, and examples repair the damage slowly.

Limitations and Future Work. Our claims cover Bayes-optimal predictors and small meta-trained Transformers in a linear-Gaussian setting, chosen for exact calculation; whether they carry to large language models reading text is open. The ESS scores one-step prediction: Definition 1 counts the examples already in the prompt, not a sequential session in which the predictor’s own feedback accumulates. Over such horizons a reliable description’s long-horizon ESS tends to

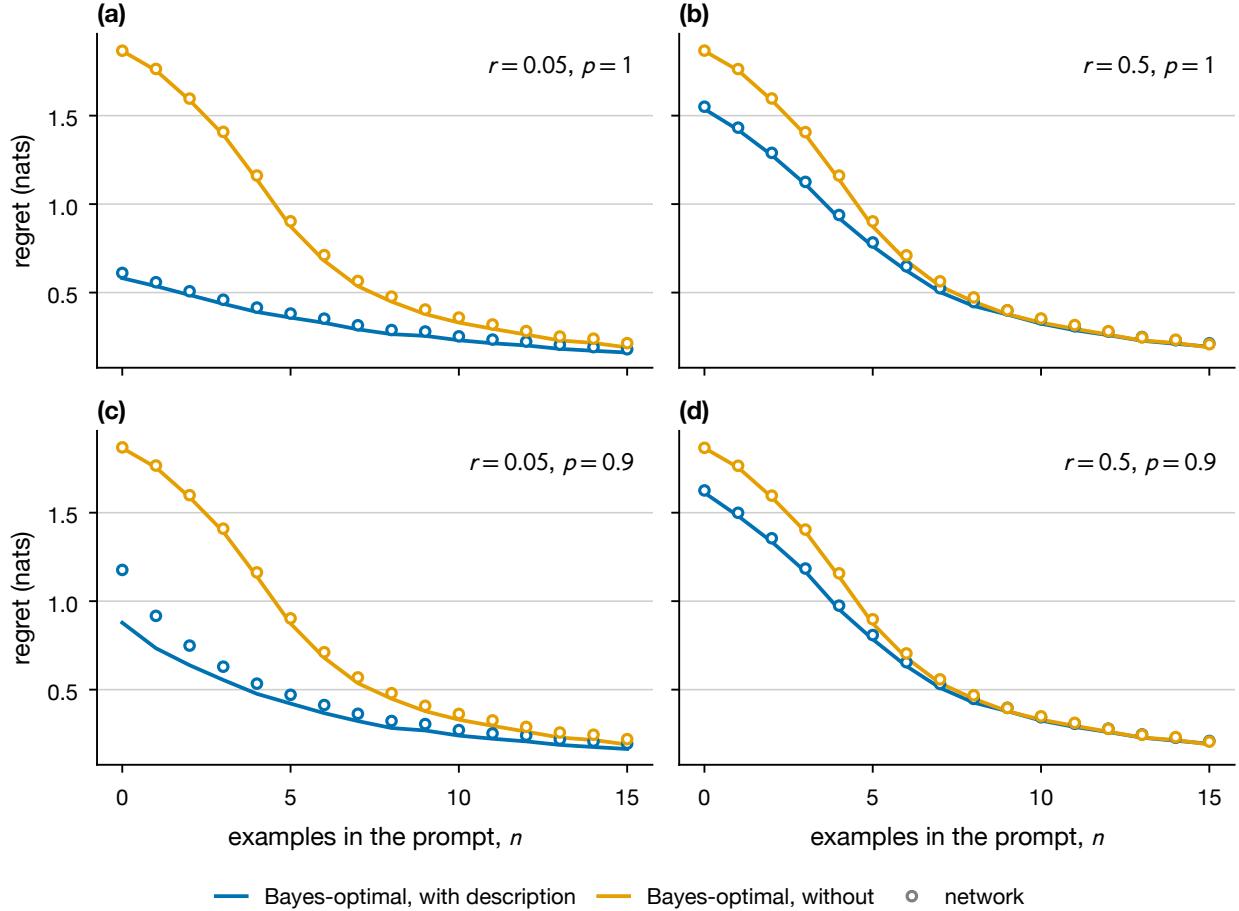


Figure 3: **Networks trained on the matching distribution track the Bayes-optimal predictor, except the precise unreliable descriptor.** Regret against the number of examples for the network (markers; standard errors over 20,000 prompts are smaller than the markers) and the Bayes-optimal predictor (lines); one trained model per panel. The lower curve in each panel is prompts with a descriptor followed by n examples, the upper curve prompts with n examples alone. The ESS is the n at which the upper curve reaches the lower curve’s value at $n = 0$. (a) $r = 0.05, p = 1$; (b) $r = 0.5, p = 1$; (c) $r = 0.05, p = 0.9$; (d) $r = 0.5, p = 0.9$. In (c) the network’s regret with the descriptor alone exceeds the Bayes-optimal predictor’s by 0.3 nats, and its ESS is 3.9 against 5.0.

the number of examples whose expected information gain matches its $\frac{d}{2} \log(1/r)$ nats, below the one-step value (7.8 against 14.9 at $d = 16, a_0 = 10, r = 0.1$); the horizon dependence of unreliable descriptions is left for future work.

References

- Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.
- Tom B. Brown et al. Language models are few-shot learners. In *NeurIPS*, 2020.
- Nicolò Cesa-Bianchi and Gábor Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.
- Michael Evans and Hadas Moshonov. Checking for prior-data conflict. *Bayesian Analysis*, 1(4):893–914, 2006.
- Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What can transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

- Tim Genewein, Li Kevin Wenliang, Jordi Grau-Moya, Anian Ruoss, Laurent Orseau, and Marcus Hutter. Understanding prompt tuning and in-context learning via meta-learning. In *NeurIPS*, 2025.
- Ritwik Gupta, Rodolfo Corona, Jiaxin Ge, Eric Wang, Dan Klein, Trevor Darrell, and David M. Chan. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.
- Ukyo Honda, Soichiro Murakami, and Peinan Zhang. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.
- Ruomin Huang and Rong Ge. Task descriptors help transformers learn linear models in-context. In *ICLR*, 2025.
- Teven Le Scao and Alexander M. Rush. How many data points is a prompt worth? In *NAACL*, 2021.
- Ziqian Lin and Kangwook Lee. Dual operating modes of in-context learning. In *ICML*, 2024.
- Ziqian Lin, Shubham Kumar Bharti, and Kangwook Lee. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.
- Neri Merhav and Meir Feder. Universal prediction. *IEEE Transactions on Information Theory*, 44(6):2124–2147, 1998.
- Satoshi Morita, Peter F. Thall, and Peter Müller. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.
- Robb J. Muirhead. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.
- Beat Neuenschwander, Sebastian Weber, Heinz Schmidli, and Anthony O’Hagan. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.
- Madhur Panwar, Kabir Ahuja, and Navin Goyal. In-context learning through the Bayesian prism. In *ICLR*, 2024.
- Allan Raventós, Mansheej Paul, Feng Chen, and Surya Ganguli. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.
- Matthew Reimherr, Xiao-Li Meng, and Dan L. Nicolae. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society: Series B*, 83(3):413–437, 2021.
- Yuriy A. Reznik. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.
- Heinz Schmidli, Sandro Gsteiger, Satrajit Roychoudhury, Anthony O’Hagan, David Spiegelhalter, and Beat Neuenschwander. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.
- X. Tong, Y. Zeng, and J. Zhang. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.
- Dietrich von Rosen. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.
- Albert Webson and Ellie Pavlick. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.
- Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.
- Qingyang Zhu, Eric Karl Oermann, and Kyunghyun Cho. Multi-task Bayesian in-context learning. In *ICML*, 2026. *arXiv:2606.20538*.

Table 2: Exact ESS of a reliable description ($p = 1$) for $a_0 = 10$, three seeds (standard deviation across seeds below 2% of each value), against the additive rule $(d - 1)(1 - r) + 1/(ra_0)$ of Proposition 1, which is within 2.2 examples of every cell.

d	r	ESS	$(d - 1)(1 - r) + 1/(ra_0)$
4	0.1	4.41	3.7
4	0.01	14.5	13
4	0.001	105	103
16	0.1	14.9	14.5
16	0.01	26.1	24.9
16	0.001	117	115
64	0.1	57.7	57.7
64	0.01	73.3	72.4
64	0.001	164	163

Table 3: ESS of unreliable descriptions ($d = 16$, $a_0 = 10$), three seeds; standard deviation across seeds at most 2% of each value. Down a column precision rises tenfold per row; across a row the description is wrong never, one time in a hundred, and one time in ten.

r	$p = 1$	$p = 0.99$	$p = 0.9$
0.1	14.9	14.6	13.1
0.01	26.1	24.2	18.4
0.001	116	64.2	23.1

A Exact values

Tables 2 and 3 give the exact one-step ESS behind §4 and §5 at the ends of the grids; Figure 2 shows the relationships between them. The two tables are separate simulations, so the one quantity they share, the reliable description at $d = 16$, $r = 0.001$, differs in its last digit: 117 against 116.

B Proof of Proposition 1

Proof. Throughout, $\varepsilon = 1/a_0$, x is the fresh query with $A = \|x\|^2 \sim \chi_d^2$, and $\Sigma = (\varepsilon I + X_n^\top X_n)^{-1}$. We use $\mathbb{E} \log \chi_k^2 = \psi(k/2) + \log 2$, $\mathbb{E}[1/\chi_k^2] = 1/(k - 2)$, $\mathbb{E}[1/\chi_k^4] = 1/((k - 2)(k - 4))$, and the Wishart facts that for $W \sim W_q(p, I)$ with $p \geq q$ and $z \sim \mathcal{N}(0, I_q)$ independent of W , $z^\top W^{-1} z$ is distributed as $\chi_q^2 / \chi_{p-q+1}^2$ with independent numerator and denominator (Muirhead, 1982, Theorem 3.2.12), $\mathbb{E} W^{-1} = I/(p - q - 1)$, and $\mathbb{E} \text{tr} W^{-2} = q(p - 1)/((p - q)(p - q - 1)(p - q - 3))$ (von Rosen, 1988).

Step 1: regret is leftover variance. The Bayes predictive for y is the true conditional law of y given the data, a Gaussian with variance $1 + x^\top \Sigma x$. The expected log loss of a true conditional law is its entropy, $\frac{1}{2} \log(2\pi e(1 + x^\top \Sigma x))$; the oracle’s predictive is $\mathcal{N}(w^\top x, 1)$ with entropy $\frac{1}{2} \log(2\pi e)$. So $R^{\text{ex}}(n) = \frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$, and since Σ shrinks pathwise as n grows, R^{ex} is nonincreasing in n . For the description, $n = 0$ and $\bar{\Sigma} = ra_0 I$, so with $g(t) := \mathbb{E} \log(1 + At)$,

$$2R^{\text{desc}} = g(1/c) = \log(2ra_0) + \psi(d/2) + \delta^{\text{desc}}, \quad \delta^{\text{desc}} = \mathbb{E} \log(1 + c/A) = \frac{c}{d-2} + O\left(\frac{c^2}{d^2}\right),$$

the last from $y - y^2/2 \leq \log(1 + y) \leq y$ with $y = c/A$ ($d \geq 5$).

Step 2: fewer than d examples. Write $X_n^\top X_n = \sum_{i \leq n} \lambda_i u_i u_i^\top$ with $\lambda_i > 0$ almost surely, let $P_\perp$ project onto the $k = d - n$ directions the examples did not look at, and $Q = \|P_\perp x\|^2 \sim \chi_k^2$. Then Σ has eigenvalue a_0 on those directions and $1/(\varepsilon + \lambda_i)$ on the others, so

$$1 + x^\top \Sigma x = a_0 Q + T, \quad T = 1 + V, \quad V = \sum_{i \leq n} \frac{z_i^2}{\varepsilon + \lambda_i}, \quad z_i = u_i^\top x.$$

Given X_n , the z_i are i.i.d. standard normal and independent of Q , and the λ_i are the eigenvalues of $X_n X_n^\top \sim W_n(d, I)$. Hence

$$2R^{\text{ex}}(n) = \log(2a_0) + \psi(k/2) + \delta^{\text{ex}}, \quad \delta^{\text{ex}} = \mathbb{E} \log \left(1 + \frac{\varepsilon T}{Q} \right) \geq 0,$$

which is the lower bound in (i); $\varepsilon T \rightarrow 0$ as $a_0 \rightarrow \infty$, so $\delta^{\text{ex}} \rightarrow 0$ by monotone convergence, which is the equality.

For the finite- a_0 term, let $F = \sum_i z_i^2 / \lambda_i = z^\top (X_n X_n^\top)^{-1} z \sim \chi_n^2 / \chi_{k+1}^2$, so $\mathbb{E}F = n/(k-1)$ and $\mathbb{E}F^2 = n(n+2)/((k-1)(k-3))$. Then $F - \varepsilon \sum_i z_i^2 / \lambda_i^2 \leq V \leq F$, and $\mathbb{E} \sum_i z_i^2 / \lambda_i^2 = \mathbb{E} \text{tr}(X_n X_n^\top)^{-2} = n(d-1)/(k(k-1)(k-3))$. Applying $y - y^2/2 \leq \log(1+y) \leq y$ with $y = \varepsilon T/Q$ and the independence of Q from T ,

$$\frac{\varepsilon(1 + \mathbb{E}V)}{k-2} - \frac{\varepsilon^2 \mathbb{E}T^2}{2(k-2)(k-4)} \leq \delta^{\text{ex}} \leq \frac{\varepsilon(1 + \mathbb{E}F)}{k-2}, \quad \text{so} \quad \delta^{\text{ex}} = \frac{\varepsilon(d-1)}{(k-1)(k-2)} + O\left(\frac{\varepsilon^2 d^2}{k^4}\right), \quad k \geq 5.$$

Step 3: solve (i). Matching $R^{\text{ex}}(n) = R^{\text{desc}}$ cancels $\log a_0$:

$$\psi(k/2) - \psi(d/2) = \log r + \delta^{\text{desc}} - \delta^{\text{ex}}.$$

Read it as in the high-SNR case: the k untouched directions must be as big, on a log scale, as d directions each shrunk by r , with the finite-SNR leftovers δ^{desc} and δ^{ex} as corrections. Treat k as real; the paper's ESS interpolates R^{ex} linearly between integers, and since the right side of the display is, as a function of n , smooth with second derivative $O(1/k^2)$ and slope of order $1/k$, that moves the root by $O(1/k)$, which the error term absorbs. With $\psi(z) = \log z - 1/(2z) + O(z^{-2})$,

$$k = rd \exp(\eta), \quad \eta = \frac{1}{k} - \frac{1}{d} + \delta^{\text{desc}} - \delta^{\text{ex}} + O(k^{-2}).$$

As $rd \rightarrow \infty$ every term of η vanishes, so $k = rd(1 + o(1))$ as in the high-SNR proof, and then $1/k = O(1/(rd))$, $\delta^{\text{desc}} = O(c/d)$, and $\delta^{\text{ex}} = O(\varepsilon d/k^2) = O(c/(rd))$ because $\varepsilon = rc$. So $\eta = O((1+c)/(rd))$ and $k = rd + rd\eta + O((1+c)^2/(rd))$. In $rd\eta$: $rd/k = 1 + O((1+c)/(rd))$; $rd\delta^{\text{desc}} = rc + O(rc(1+c)/d) = 1/a_0 + O((1+c)^2/(rd))$; and $rd\delta^{\text{ex}} = (\varepsilon/r)(rd/k)^2(1 + O(1/k)) + O(\varepsilon^2 rd^3/k^4) = c + O(c(1+c)/(rd)) + O(c^2/(rd))$. Hence

$$k = rd + 1 - r + \frac{1}{a_0} - c + O\left(\frac{(1+c)^2}{rd}\right), \quad \text{ESS} = d - k = (d-1)(1-r) + c - \frac{1}{a_0} + O\left(\frac{(1+c)^2}{rd}\right),$$

and $c - 1/a_0 = c - rc = (1-r)c$. For $c \leq 1$ the error is $O(1/(rd))$, which is (i). As $a_0 \rightarrow \infty$ with r fixed, $c \rightarrow 0$ and this is the high-SNR rule $(d-1)(1-r) + O(1/(rd))$.

Step 4: at least d examples, an exact representation. Rotate coordinates so that $x = \|x\|e_1$; the rows of the rotated X_n are still i.i.d. $\mathcal{N}(0, I_d)$ and independent of $\|x\|$. Then $x^\top \Sigma x = A \Sigma_{11} = A/S$, where by the Schur complement and the Woodbury identity, with X_1 the first column of X_n and X_{-1} the other $d-1$ columns,

$$S = \varepsilon + X_1^\top (I_n + a_0 X_{-1} X_{-1}^\top)^{-1} X_1.$$

Given X_{-1} , the matrix in the middle has eigenvalue 1 on the $m = n - d + 1$ directions orthogonal to the columns of X_{-1} and $1/(1 + a_0 \mu_j)$ on the other $d-1$, where μ_j are the eigenvalues of $M = X_{-1}^\top X_{-1} \sim W_{d-1}(n, I)$. Since $X_1 \sim \mathcal{N}(0, I_n)$ is independent of X_{-1} , its coordinates in that eigenbasis are i.i.d. standard normal, and

$$S = \varepsilon + Q + \sum_{j < d} \frac{z_j^2}{1 + a_0 \mu_j}, \quad Q \sim \chi_m^2, \quad \text{so} \quad Q \leq S \leq Q + \varepsilon(1 + F), \quad F = z^\top M^{-1} z \sim \frac{\chi_{d-1}^2}{\chi_{n-d+2}^2},$$

with A, Q , and (z, M) independent, $\mathbb{E}F = (d-1)/(n-d)$ and $\mathbb{E}F^2 = (d-1)(d+1)/((n-d)(n-d-2))$.

At $a_0 = \infty$, $S = Q$ and $1 + A/Q = (A+Q)/Q$ with $A+Q \sim \chi_{n+1}^2$, so $\mathbb{E} \log(1 + A/Q) = \psi(\frac{n+1}{2}) - \psi(\frac{m}{2}) = 2\Phi(n)$. Since $S \geq Q$ for every a_0 , $R^{\text{ex}}(n) \leq \Phi(n)$, with equality as $a_0 \rightarrow \infty$ by monotone convergence: the first claim of (ii).

Step 5: the universal bound. Conditionally on A , $\log(1 + A/Q)$ is concave in $1/Q$, so Jensen gives $2\Phi(n) \leq \mathbb{E} \log(1 + A/(m-2)) = g(1/(m-2))$ for $m > 2$. If $m-2 \geq c$, that is $n \geq c + d + 1$, then $g(1/(m-2)) \leq g(1/c) = 2R^{\text{desc}}$, so $R^{\text{ex}}(n) \leq \Phi(n) \leq R^{\text{desc}}$. The smallest such integer is at most $c + d + 2$, and R^{ex} is nonincreasing, so $\text{ESS} \leq c + d + 2$ in every setting.

Step 6: solve (ii). Two sandwiches. First, finite a_0 : with $D = \log(1 + A/Q) - \log(1 + A/S) = \log\left(1 + \frac{A(S-Q)}{Q(S+A)}\right) \geq 0$, the bounds $0 \leq S - Q \leq \varepsilon(1 + F)$, $1/(S + A) \geq 1/Q - (S + A - Q)/Q^2$, and $y - y^2/2 \leq \log(1 + y) \leq y$ give, for $a_0 \geq 1$ and $n \geq d + 6$,

$$\mathbb{E}D = \frac{\varepsilon d}{m^2} \left(1 + O\left(\frac{d+1}{m}\right)\right), \quad \text{so} \quad 2R^{\text{ex}}(n) = 2\Phi(n) - \frac{\varepsilon d}{m^2} + O\left(\frac{\varepsilon d^2}{m^3}\right).$$

(The upper bound is $\mathbb{E}[\varepsilon A(1 + F)/Q^2] = \varepsilon d(n - 1)/((n - d)(n - d - 1)(n - d - 3))$; the lower bound subtracts $\mathbb{E}[\varepsilon A(\varepsilon(1 + F) + A)/Q^3]$ and $\frac{1}{2}\mathbb{E}[\varepsilon^2 A^2(1 + F)^2/Q^4]$, both $O(\varepsilon d^2/m^3)$.) Second, Φ against its mean: $g''(t) = -\mathbb{E}[A^2/(1 + At)^2] \geq -(d^2 + 2d)$ and $\text{Var}(1/Q) = 2/((m - 2)^2(m - 4))$, so a second-order Taylor bound around $\mathbb{E}[1/Q] = 1/(m - 2)$ gives

$$2\Phi(n) = g\left(\frac{1}{m-2}\right) - \vartheta \frac{d^2 + 2d}{(m - 2)^2(m - 4)}, \quad \vartheta \in [0, 1].$$

Together,

$$2R^{\text{ex}}(n) - 2R^{\text{desc}} = g\left(\frac{1}{m-2}\right) - g\left(\frac{1}{c}\right) - \frac{\varepsilon d}{m^2} + O\left(\frac{d^2}{m^3}\right).$$

Now $d(1 - (d + 2)t) \leq g'(t) \leq d$ because $A - A^2t \leq A/(1 + At) \leq A$. Put $m - 2 = c + s$ with $|s| \leq c/2$. Then $g(1/(m - 2)) - g(1/c) = -\frac{ds}{c(c+s)}(1 + O(d/c))$ and $\varepsilon d/m^2 = \frac{\varepsilon d}{c^2}(1 + O((|s| + 1)/c))$, so

$$2R^{\text{ex}}(n) - 2R^{\text{desc}} = -\frac{d}{c^2} \left[s + \varepsilon + O\left(\frac{(1 + |s|)(d + |s|)}{c}\right) \right].$$

At $s = -\varepsilon \pm Kd/c$ the bracket has the sign $\pm$ once K exceeds the constant in the O (for c/d large, $|s| = O(1)$). So $R^{\text{ex}} - R^{\text{desc}}$ changes sign between $n = c + d + 1 - \varepsilon - Kd/c$ and $n = c + d + 1 - \varepsilon + Kd/c$; linear interpolation between integers moves the crossing by $O(1/c)$, and the moment conditions $n \geq d + 6$ hold since $n \geq c$. Hence $\text{ESS} = c + d + 1 - 1/a_0 + O(d/c)$, which is (ii). $\square$

C Proof of Proposition 2

Proof. Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ be everything the predictor sees before predicting y , and $Z = 1$ if the description is correct, $Z = 0$ otherwise.

Step 1: the predictive is a π_n -mixture. The Bayes-optimal predictive is $q(y) = \pi_n f_1(y) + (1 - \pi_n) f_0(y)$, where $\pi_n = P(Z = 1 \mid D_n)$ and f_1, f_0 are the predictives of the two components given D_n : f_1 is the reliable-description predictor's, and f_0 is the examples-only predictor's, because under $Z = 0$ the description is independent of w and carries no information about y . At $n = 0$, $\pi_0 = p$: seeing m says nothing about Z , because m has the same law $\mathcal{N}(0, (1 - r)a_0 I)$ either way, and x is independent of Z , so there is nothing to update on. The components are then $f_1 = \mathcal{N}(m^\top x, 1 + ra_0 \|x\|^2)$ and $f_0 = \mathcal{N}(0, 1 + a_0 \|x\|^2)$.

Step 2: the twin. Given (D_n, Z) , the true conditional density of y is f_Z . So a twin who is told Z predicts with f_1 when $Z = 1$, with regret $R_{p=1}^{\text{desc}}(n)$, and with f_0 when $Z = 0$, with regret $R^{\text{ex}}(n)$. Its regret is $p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n)$.

Step 3: the learner cannot beat the twin (lower bound). Given (D_n, Z) , the expected log loss of any density q exceeds that of the true density f_Z by $\mathbb{E}[-\log q \mid D_n, Z] - \mathbb{E}[-\log f_Z \mid D_n, Z] = \text{KL}(f_Z \parallel q) \geq 0$. Averaging over Z and D_n gives $R^{\text{desc}}(n) \geq p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n)$.

Step 4: the learner loses at most the identification term (upper bound). Pointwise, $q \geq \pi_n f_1$ and $q \geq (1 - \pi_n) f_0$, so $-\log q \leq -\log f_Z - \log \pi_n(Z)$ with $\pi_n(1) = \pi_n$ and $\pi_n(0) = 1 - \pi_n$. Taking expectations gives the upper bound with the identification term $\mathbb{E}[-\log \pi_n(Z)]$. At $n = 0$ it is $p \log(1/p) + (1 - p) \log(1/(1 - p)) = H(p)$.

Step 5: the identification term is non-increasing. $\mathbb{E}[-\log \pi_n(Z)] = H(Z \mid D_n)$, the conditional entropy of Z given the data. The query x is independent of Z and of everything else in D_n , so $H(Z \mid D_n) = H(Z \mid x_{1:n}, y_{1:n}, m)$, and these conditioning sets are nested in n . Conditioning on more does not increase conditional entropy, so the term is non-increasing.

Step 6: the $n = 0$ case. $R_{p=1}^{\text{desc}}(0) = \frac{1}{2} \mathbb{E} \log(1 + ra_0 \|x\|^2)$ is nonnegative, which gives the floor $(1 - p) R^{\text{ex}}(0)$ for every r . The ESS is the n at which $R^{\text{ex}}(n) = R^{\text{desc}}(0) \geq (1 - p) R^{\text{ex}}(0)$, and $R^{\text{ex}}(n)$ is non-increasing in n (a

predictor given $n + 1$ examples can ignore one, and Bayes-optimal regret cannot rise with more information), so the ESS cannot exceed the n at which $R^{\text{ex}}(n) = (1 - p) R^{\text{ex}}(0)$. Finally $R_{p=1}^{\text{desc}}(0)$ decreases to 0 as $r \rightarrow 0$ by monotone convergence, which gives the two limits. $\square$